

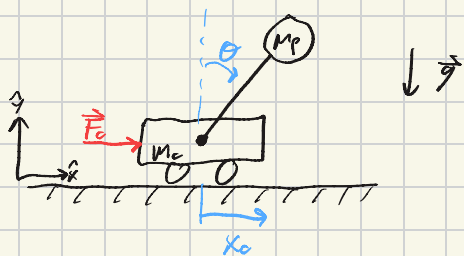
## Last Time

- Generalized Coordinates
- Work
- d'Alembert's Principle
- Euler-Lagrange Equation

## Today

- Cart Pole
- Manipulator Equation
- Mass Matrix
- Rotating Frames

## Cart Pole



- Use coordinates  $x_c$  for cart position and  $\theta$  for pendulum angle
- We'll also use  $x_p, y_p$  Cartesian coordinates for the pendulum to help calculate energy:

$$x_p = x_c + l \sin(\theta) \Rightarrow \dot{x}_p = \dot{x}_c + l \cos(\theta) \dot{\theta}$$

$$y_p = l \cos(\theta) \Rightarrow \dot{y}_p = -l \sin(\theta) \dot{\theta}$$

- Potential energy:

$$U = m g y_p = m g l \cos(\theta)$$

- Kinetic Energy:

$$\begin{aligned} T &= \frac{1}{2} m_c \dot{x}_c^2 + \frac{1}{2} m_p (\dot{x}_p^2 + \dot{y}_p^2) \\ &= \frac{1}{2} m_c \dot{x}_c^2 + \frac{1}{2} m_p (\dot{x}_c + l \cos(\theta) \dot{\theta})^2 + \frac{1}{2} m_p l^2 \sin^2(\theta) \dot{\theta}^2 \\ &= \frac{1}{2} (m_c + m_p) \dot{x}_c^2 + \frac{1}{2} m_p l^2 \dot{\theta}^2 + m_p l \cos(\theta) \dot{\theta} \dot{x}_c \end{aligned}$$

- Lagrangian:

$$\begin{aligned} L &= T - U \\ &= \frac{1}{2} (m_c + m_p) \dot{x}_c^2 + \frac{1}{2} m_p l^2 \dot{\theta}^2 + m_p l \cos(\theta) \dot{\theta} \dot{x}_c - m_p g l \cos(\theta) \end{aligned}$$

- Euler-Lagrange Equation:

$$\frac{d}{dt} \left( \frac{\partial L}{\partial \dot{q}} \right) - \frac{\partial L}{\partial q} = Q, \quad q = \begin{bmatrix} x_c \\ \theta \end{bmatrix}, \quad Q = \begin{bmatrix} F_c \\ \tau_p \end{bmatrix}$$

- We can do this one component at a time:

$$\frac{d}{dt} \left( \frac{\partial L}{\partial \dot{x}_c} \right) - \frac{\partial L}{\partial x_c} = F_c$$

$$\Rightarrow (m_c + m_p) \ddot{x}_c + m_p l \cos(\theta) \ddot{\theta} - m_p l \sin(\theta) \dot{\theta}^2 = F_c$$

$$\frac{d}{dt} \left( \frac{\partial L}{\partial \dot{\theta}} \right) - \frac{\partial L}{\partial \theta} = 0$$

$$\Rightarrow m_p l \cos(\theta) \ddot{x}_c + m_p l^2 \ddot{\theta} - m_p g \sin(\theta) = 0$$

## Manipulator Equation

- We can group terms in the CDEs above:

$$\underbrace{\begin{bmatrix} (m_c + m_p) & m_p l \cos(\theta) \\ m_p l \cos(\theta) & m_p l^2 \end{bmatrix}}_{\substack{M(q) \\ \text{"mass matrix"}}} \underbrace{\begin{bmatrix} \ddot{x}_c \\ \ddot{\theta} \end{bmatrix}}_{\dot{q}} + \underbrace{\begin{bmatrix} m_p l \sin(\theta) \dot{\theta} \\ 0 \end{bmatrix}}_{C(q, \dot{q}) \text{ "Coriolis"}} + \underbrace{\begin{bmatrix} 0 \\ -m_p g l \sin(\theta) \end{bmatrix}}_{g(q) \text{ "gravity" / potential term}} = \underbrace{\begin{bmatrix} F_c \\ 0 \end{bmatrix}}_Q$$

- To calculate acceleration:

$$\ddot{q} = M(q)^{-1} [Q - C(q, \dot{q}) - g(q)]$$

- Solving a linear system ( $Ax=b$ ) is  $O(n^3)$  (takes time proportional to  $n = \dim x$  cubed).

$\Rightarrow$  In general computing dynamics is  $O(n^3)$  where  $n = \text{number of DOFs}$

- Many software packages compute/solve the manipulator equation.

- Sometimes  $g(q)$  is folded into  $C(q, \dot{q})$ . The term is then called "dynamic bias".

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## Mass Matrix

- Kinetic energy can be written  $T = \frac{1}{2} \dot{q}^T M(q) \dot{q}$

- Since  $T \geq 0 \Rightarrow M(q)$  is symmetric positive-definite ( $M(q) \geq 0$ )

$\Rightarrow$  All eigenvalues are real and positive

- If system is a set of point masses  $m_1, \dots, m_n$  with Cartesian positions  $r_1, \dots, r_n$

$$r(q) = \begin{bmatrix} x_c \\ y_c \\ x_p \\ y_p \end{bmatrix} = \begin{bmatrix} x_c \\ 0 \\ x_c + l \sin(\theta) \\ l \cos(\theta) \end{bmatrix} \Rightarrow \dot{r} = \begin{bmatrix} \dot{x}_c \\ 0 \\ \dot{x}_c + l \cos(\theta) \dot{\theta} \\ -l \sin(\theta) \dot{\theta} \end{bmatrix}$$

$$\dot{r}(q) = \underbrace{\frac{\partial r}{\partial q}}_{J(q)} \dot{q} = \underbrace{\begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 1 & l \cos(\theta) \\ 0 & -l \sin(\theta) \end{bmatrix}}_{J(q)} \underbrace{\begin{bmatrix} \dot{x}_c \\ \dot{\theta} \end{bmatrix}}_{\dot{q}}$$

$$T = \frac{1}{2} m_c \dot{r}_c^2 + \frac{1}{2} m_p \dot{r}_p^2 = \frac{1}{2} \begin{bmatrix} \dot{r}_c \\ \dot{r}_p \end{bmatrix}^T \underbrace{\begin{bmatrix} m_c I & 0 \\ 0 & m_p I \end{bmatrix}}_{\bar{M}} \begin{bmatrix} \dot{r}_c \\ \dot{r}_p \end{bmatrix} = \frac{1}{2} \dot{r}_c^T \bar{M} \dot{r}_c$$

$$= \frac{1}{2} \dot{q}^T \underbrace{J(q)^T \bar{M} J(q)}_{M(q)} \dot{q}$$

$$(AB)^T = B^T A^T$$

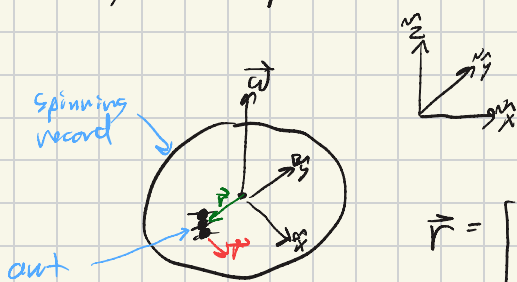
$$\Rightarrow M(q) = J(q)^T \bar{M} J(q)$$

## Rotating Reference Frames

- Shows up in many aerospace applications:

- Aircraft dynamics
- Spacecraft attitude dynamics
- Orbit mechanics (CR3BP)
- Inertial navigation

- Planar toy example:



$$\vec{r} = \begin{bmatrix} {}^N r_1 \\ {}^N r_2 \\ {}^N r_3 \end{bmatrix}^T \begin{bmatrix} \hat{x}^N \\ \hat{y}^N \\ \hat{z}^N \end{bmatrix} = \begin{bmatrix} {}^B r_1 \\ {}^B r_2 \\ {}^B r_3 \end{bmatrix}^T \begin{bmatrix} \hat{x}^B \\ \hat{y}^B \\ \hat{z}^B \end{bmatrix}$$

- As we saw before,  ${}^N r$  and  ${}^B r$  are related by a rotation matrix:

$$\underbrace{\begin{bmatrix} \hat{x}^N \\ \hat{y}^N \\ \hat{z}^N \end{bmatrix}}_I \underbrace{\begin{bmatrix} \hat{x}^B \\ \hat{y}^B \\ \hat{z}^B \end{bmatrix}^T}_{{}^N R^B} \begin{bmatrix} {}^N r_1 \\ {}^N r_2 \\ {}^N r_3 \end{bmatrix} = \underbrace{\begin{bmatrix} \hat{x}^N \\ \hat{y}^N \\ \hat{z}^N \end{bmatrix}}_{} \underbrace{\begin{bmatrix} \hat{x}^B \\ \hat{y}^B \\ \hat{z}^B \end{bmatrix}^T}_{{}^N R^B} \begin{bmatrix} {}^B r_1 \\ {}^B r_2 \\ {}^B r_3 \end{bmatrix}$$

$$\Rightarrow {}^N r = \underbrace{{}^N R^B}_{\text{function of } t \text{ because we're spinning}} {}^B r$$

- Let's say the ant is standing still on the record:

$$\Rightarrow {}^B \dot{r} = 0$$

- What is its velocity in the  $N$  frame?

\* From calculus:  $\vec{r} = {}^N R^B {}^B r \Rightarrow \dot{\vec{r}} = \dot{{}^N R^B} {}^B r + {}^N R^B \dot{{}^B r}$

\* From a little vector geometry:

$$\dot{\vec{r}} = \vec{\omega} \times \vec{r} = ({}^N R^B \vec{\omega}) \times ({}^N R^B \vec{r}) = {}^N R^B ({}^B \vec{\omega} \times {}^B \vec{r})$$

- Combine these:

$${}^N \dot{\mathbf{r}} = {}^N \dot{\mathbf{R}}^B \mathbf{q}_r = {}^N \mathbf{R}^B (\mathbf{e}_\omega \times \mathbf{e}_r)$$